

Q1. Develop a model to forecast South Korea's monthly Total Exports

Criteria:

- Forecast should have at least 3 years (36 months of datapoints) of out of sample forecast.

Rolling walk forward analysis is recommended to conserve data.

- We recommend using research papers as inspiration and a benchmark comparison.

- Following Occam's razor, the model should be explainable and any code should be easily read and run.

- We recommend using Python but do not restrict the use of other languages.

- If using Python, please use Jupyter Notebook to include brief notes about your research consisting of an introduction, methodology (description of the model/strategy you are using, optimizations, limitations, etc.), findings with visualisations and key takeaways.

Data Sources:

<https://www.ons.gov.uk/economy/nationalaccounts/balanceofpayments/timeseries/erdm/mret>

You are allowed to use other data sources.

Please fill in the nusync form for NUS Investment Society QR Department Recruitment AY26/27

Q2. Develop a trading strategy/model using an asset (or a combination of assets) from G10 FX, Gold Continuous Futures, Govt Bonds ETF/Futures provided

Criteria:

- Forecast should have at least 5 years of out of sample forecast. Rolling walk forward analysis is recommended to conserve data.
- We recommend using research papers as inspiration and a benchmark comparison.
- Following Occam's razor, the model should be explainable and any code should be easily read and run.
- We recommend using Python but do not restrict the use of other languages.
- If using Python, please use Jupyter Notebook to include brief notes about your research consisting of an introduction, methodology (description of the model/strategy you are using, optimizations, limitations, etc.), findings with visualisations and key takeaways.

Data Sources: Provided as .csv files

You are allowed to use other data sources.

Please fill in the nusync form for NUS Investment Society QR Department Recruitment AY26/27

Q3. Develop a trading strategy to trade either an asset or a combination of assets in the given global futures/ETFs

Criteria:

- We recommend exploring momentum and trend following CTA strategies, dynamic regime allocation for ETFs or volatility strategies.
- Forecast should have at least 5 years of out of sample forecast. Rolling walk forward analysis is recommended to conserve data.
- We recommend using research papers as inspiration and a benchmark comparison.
- Following Occam's razor, the model should be explainable and any code should be easily read and run.
- We recommend using Python but do not restrict the use of other languages.
- If using Python, please use Jupyter Notebook to include brief notes about your research consisting of an introduction, methodology (description of the model/strategy you are using, optimizations, limitations, etc.), findings with visualisations and key takeaways.

Data Sources:

Provided as .csv files

You are allowed to use other data sources.

Please fill in the nusync form for NUS Investment Society QR Department Recruitment AY26/27

Q4. Develop a trading strategy to trade either a digital asset or a combination of digital assets

Criteria:

- Forecast should have at least 3 months of out of sample forecast. Rolling walk forward analysis is recommended to conserve data.
- We recommend using research papers as inspiration and a benchmark comparison.
- Following Occam's razor, the model should be explainable and any code should be easily read and run.
- We recommend using Python but do not restrict the use of other languages.
- If using Python, please use Jupyter Notebook to include brief notes about your research consisting of an introduction, methodology (description of the model/strategy you are using, optimizations, limitations, etc.), findings with visualisations and key takeaways.

Data Sources:

<https://www.coingecko.com/>

Click on Coin>Historical Data>1Y>Download .csv

Feel free to use intraday data sources from Binance or other vendors.

Please fill in the nusync form for NUS Investment Society QR Department Recruitment AY26/27